

Unit 8 Pre-Test

Transformations on the Coordinate Plane

Standards assessed: NC.8.G.1–G.4

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.G.1 Identify transformations and translate figures.
• I can name translations, reflections, and rotations and translate a point.

Match each vocabulary term to its definition, then find each image point.

B ▾ translation ✓	A. A flip of a figure over a line.
A ▾ reflection ✓	B. A slide; every point moves the same distance and direction.
C ▾ rotation ✓	C. A turn of a figure about a fixed point.
D ▾ dilation ✓	D. An enlargement or reduction by a scale factor.
E ▾ congruent ✓	E. Same size and same shape.
F ▾ image ✓	F. The resulting figure after a transformation.

Coordinate rules — reference

Reflect over x-axis: $(x, y) \rightarrow (x, -y)$ · Reflect over y-axis: $(x, y) \rightarrow (-x, y)$
Rotate 180°: $(x, y) \rightarrow (-x, -y)$ · Rotate 90° CCW: $(x, y) \rightarrow (-y, x)$ · Rotate 90° CW: $(x, y) \rightarrow (y, -x)$
Dilation (center origin), scale k: $(x, y) \rightarrow (kx, ky)$

1. Name it

Which transformation is a slide?

Answer:
answer: translation

2. Name it

Which transformation is a flip?

Answer:
answer: reflection

3. Name it

Which transformation is a turn?

Answer:

answer: rotation

5. Translate

Translate (1, 5) down 2 units. New point?

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answer: (1, 3)

4. Translate

Translate (2, 3) right 4 units. New point?

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answer: (6, 3)

6. Translate

Translate (-2, 1) left 3 and up 2. New point?

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answer: (-5, 3)

SCORE 4.0 Target Content

NC.8.G.3 Reflect and rotate points on the coordinate plane.

- I can find the image of a point after a reflection or rotation.

Find each image point. Write your answer as (x, y).

7. Reflect

Reflect (4, 2) over the x-axis.

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answer: (4, -2)

8. Reflect

Reflect (3, 5) over the y-axis.

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answer: (-3, 5)

9. Reflect

Reflect (-2, 4) over the x-axis.

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answer: (-2, -4)

10. Rotate

Rotate (3, 0) 90° counterclockwise about the origin.

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answer: (0, 3)

11. Rotate

Rotate $(2, 5)$ 180° about the origin.

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answer: $(-2, -5)$

12. Reflect

Reflect $(6, -1)$ over the y-axis.

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answer: $(-6, -1)$

SCORE **5.0** Advanced Content

NC.8.G.4 Dilate figures and describe congruence and similarity.

- *I can dilate a point by a scale factor and tell whether figures are congruent or similar.*

Find each image point or answer each question.

13. Dilate

Dilate (2, 4) by scale factor 3 (center origin).

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answer: (6, 12)

15. Scale

A dilation with scale factor 2 makes the image ____ (larger or smaller).

Answer:

answer: larger

17. Similarity

After a dilation with scale factor $\neq 1$, the image is ____ to the original.

Answer:

answer: similar

14. Dilate

Dilate (10, 5) by scale factor $\frac{1}{2}$ (center origin).

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answer: (5, 2.5)

16. Congruence

After a translation, the image is ____ to the original (congruent or similar).

Answer:

answer: congruent

18. Rotate

Rotate (1, 2) 90° clockwise about the origin.

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answer: (2, -1)
